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Olarig

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(54) **HIGH-SPEED DATA TRANSFERS THROUGH
STORAGE DEVICE CONNECTORS**

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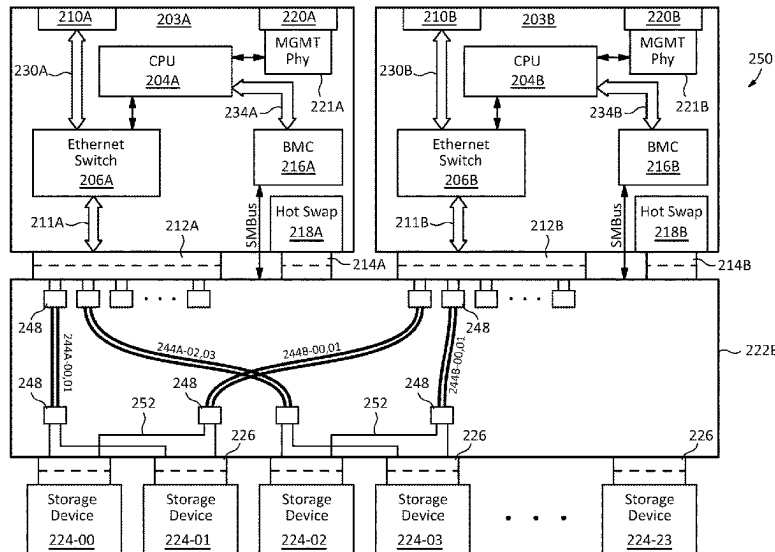
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(57) **ABSTRACT**

Data storage systems, devices and methods may use a switch
board configured to communicate using a high-speed multi-
level signaling protocol, and a midplane having one or more
multi-protocol storage device connectors configured to
couple the midplane to one or more storage devices, wherein
the midplane may be coupled to the switch board and
configured to enable the one or more storage devices to
communicate with the switch board through the one or more
multi-protocol storage device connectors using the high-
speed multi-level signaling protocol. The midplane may be
coupled to the switch board through one or more high-speed
connectors. One or more re-timers may be coupled between
one or more of the high-speed connectors and one or more
of the multi-protocol storage device connectors. One or
more cables may be used to transfer data to and from the
multi-protocol storage device connectors.

20 Claims, 12 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/924,689, filed on Oct. 22, 2019, provisional application No. 62/821,429, filed on Mar. 20, 2019.

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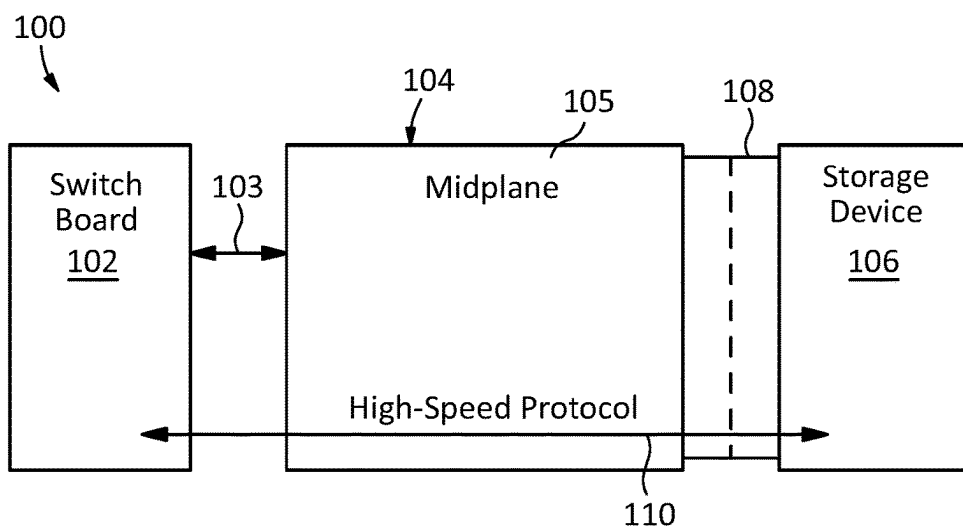


FIG. 1

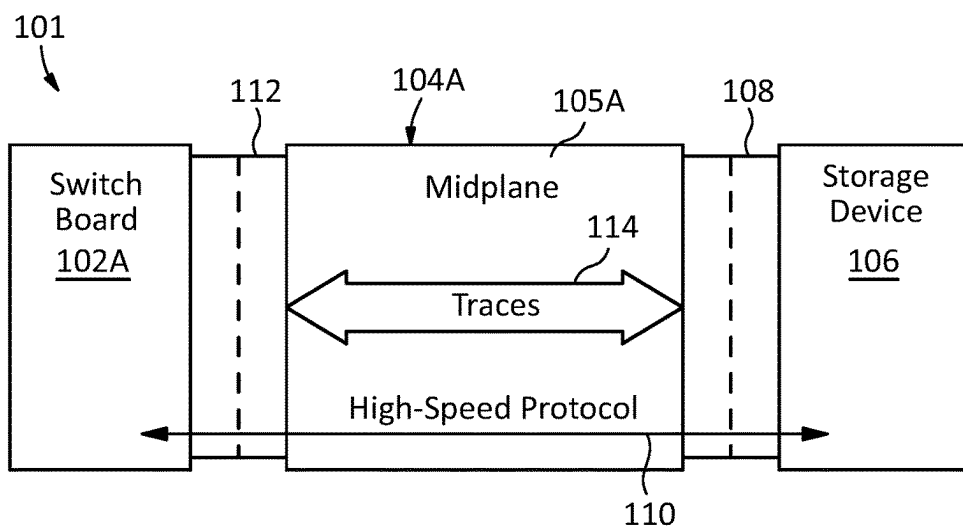


FIG. 2

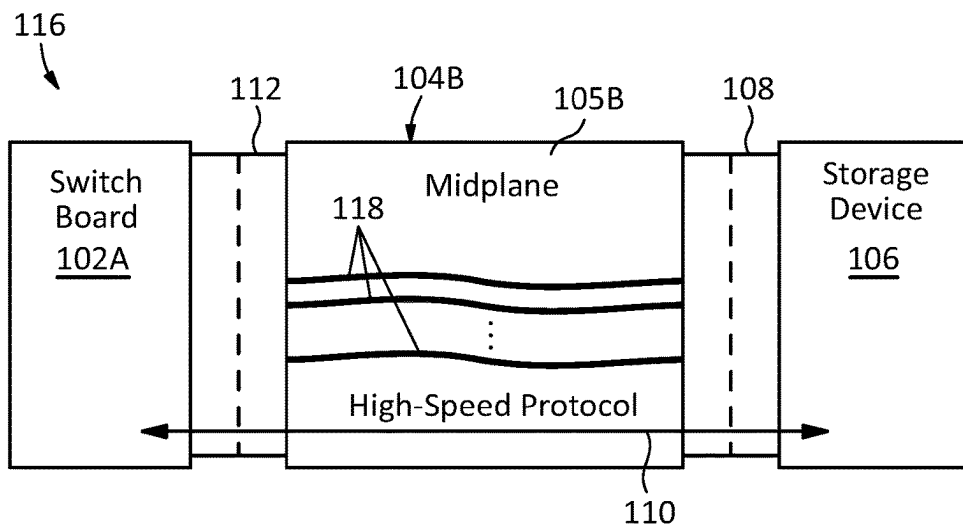


FIG. 3

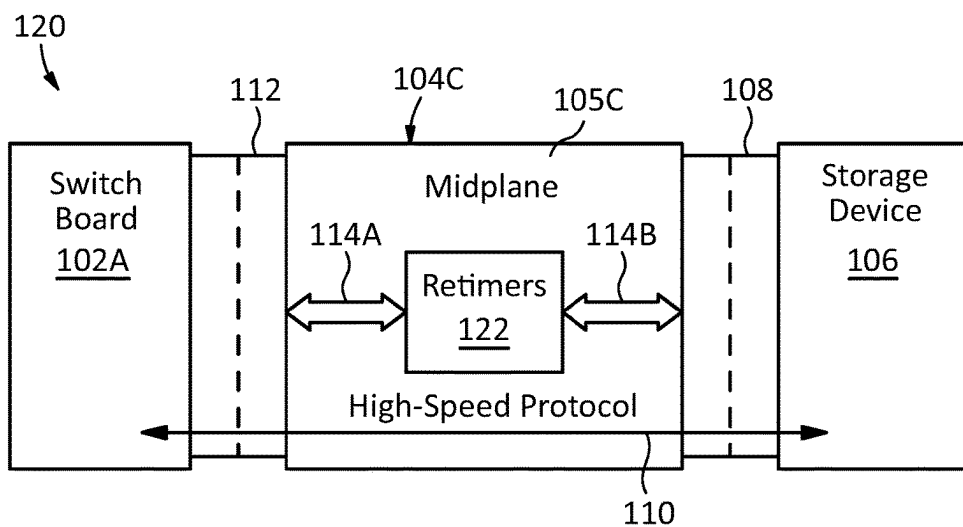


FIG. 4

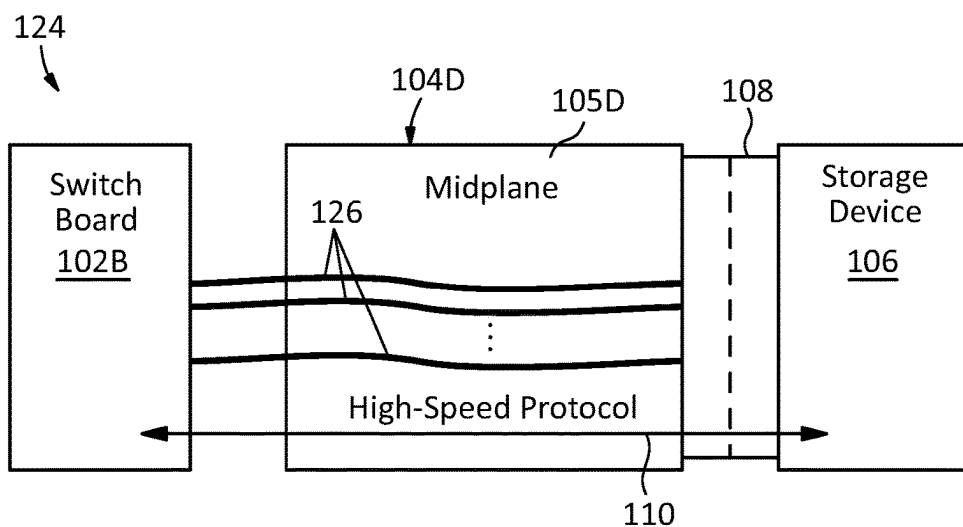


FIG. 5

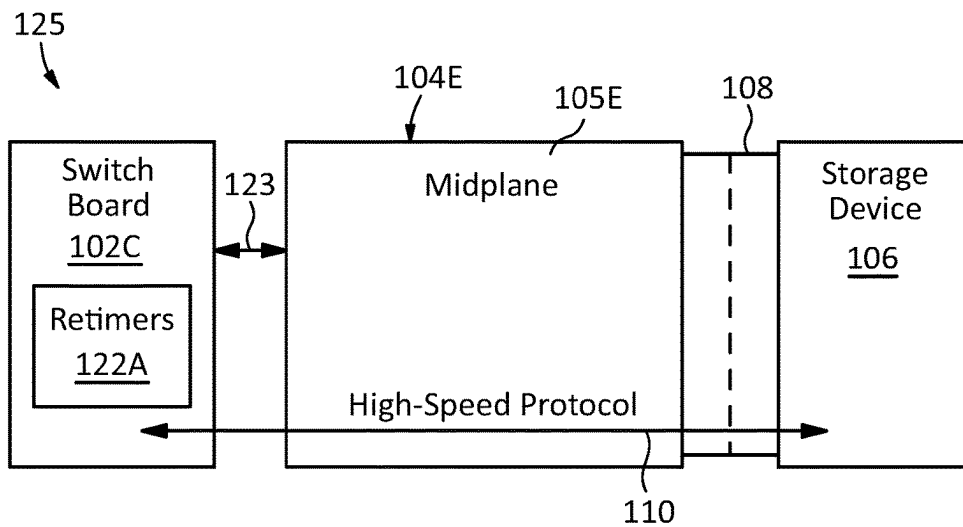


FIG. 6

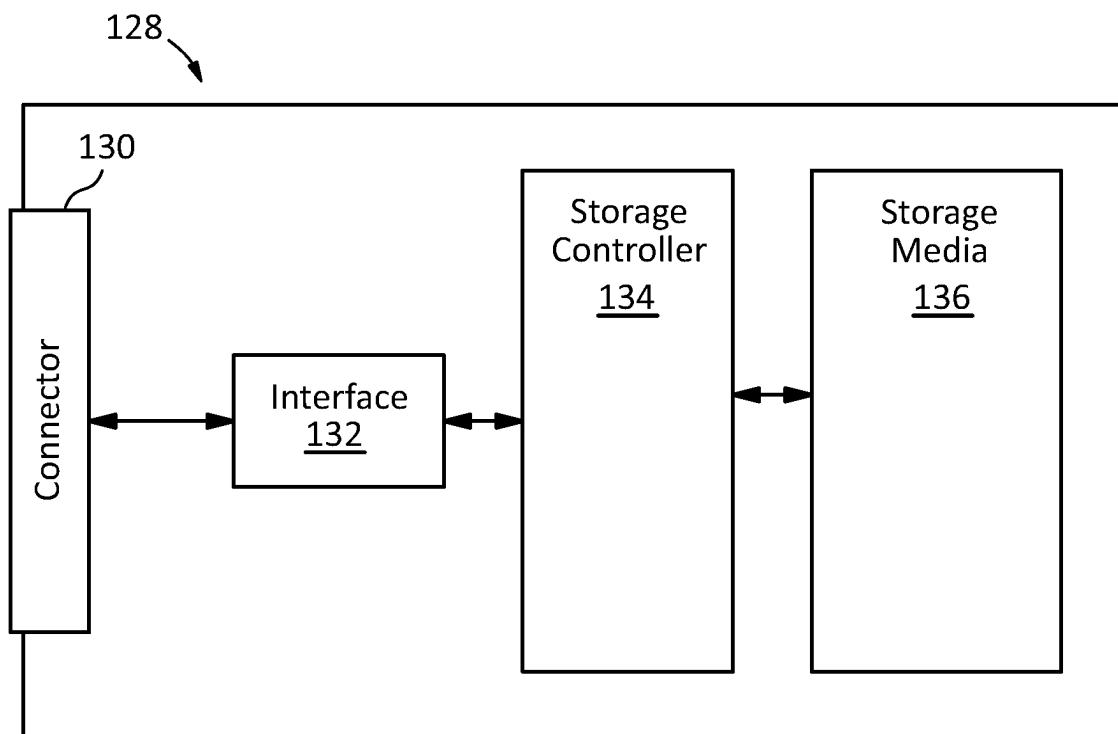


FIG. 7

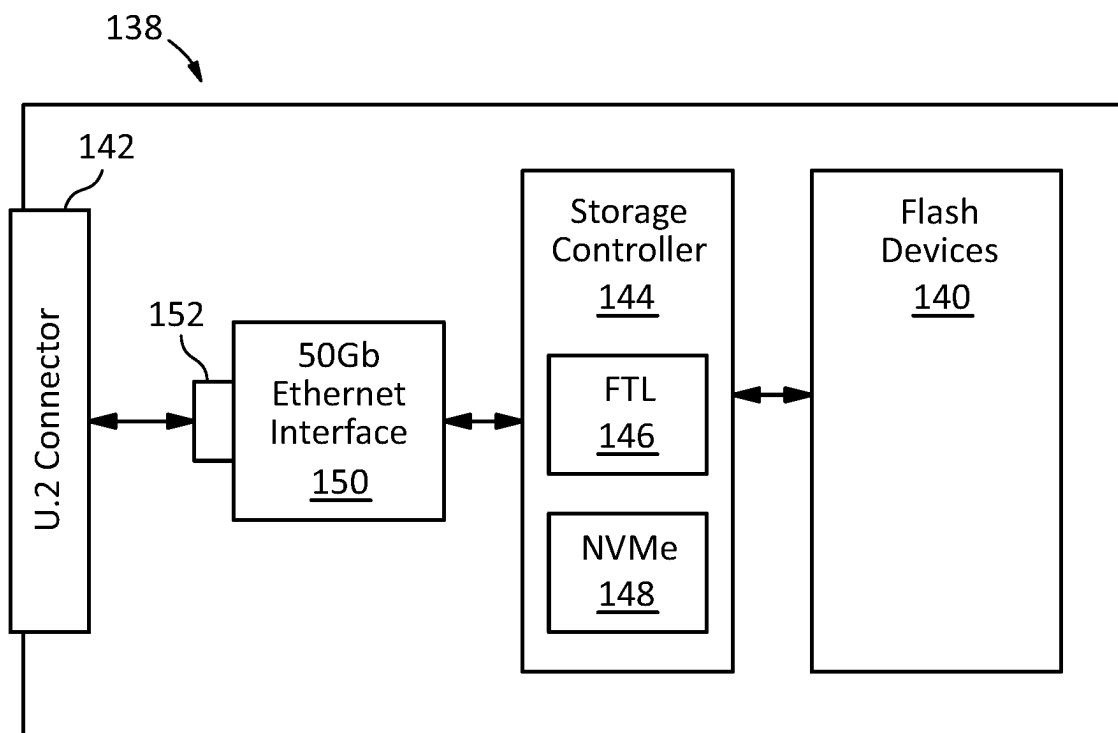


FIG. 8

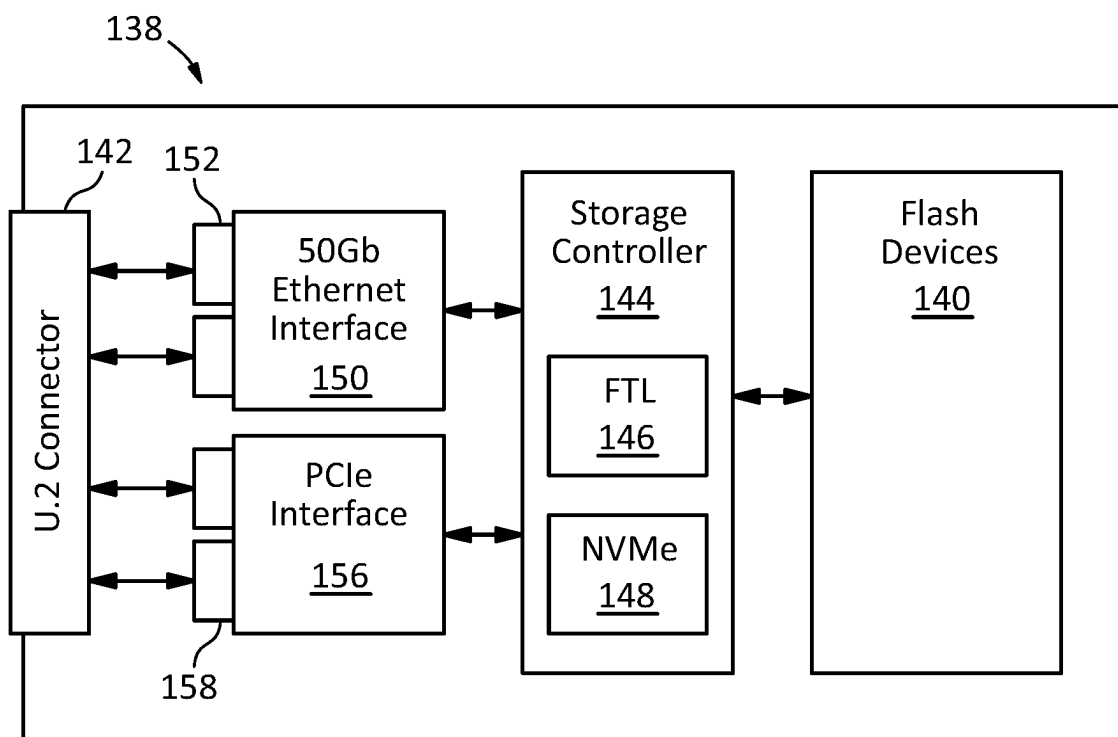


FIG. 9

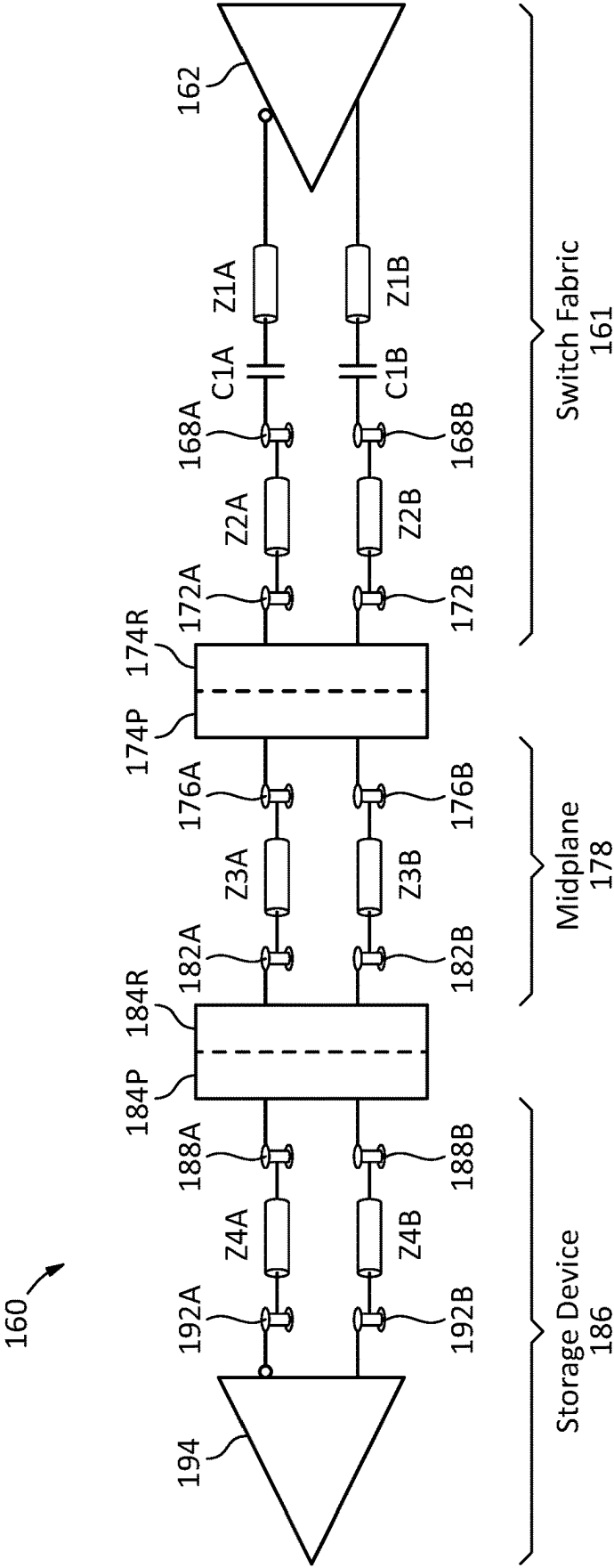


FIG. 10

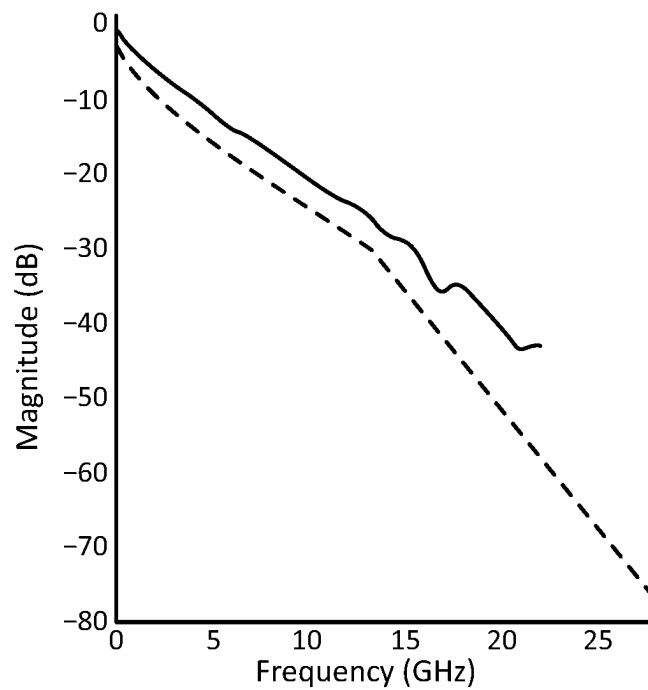


FIG. 11

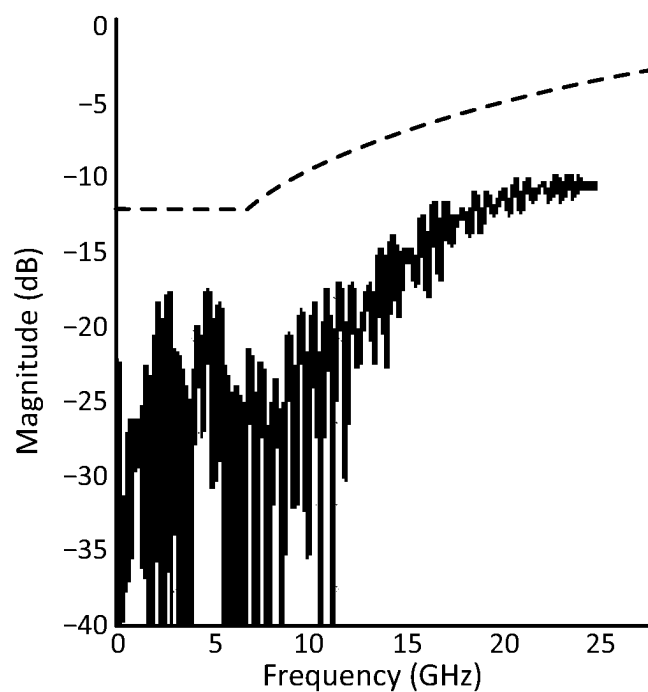
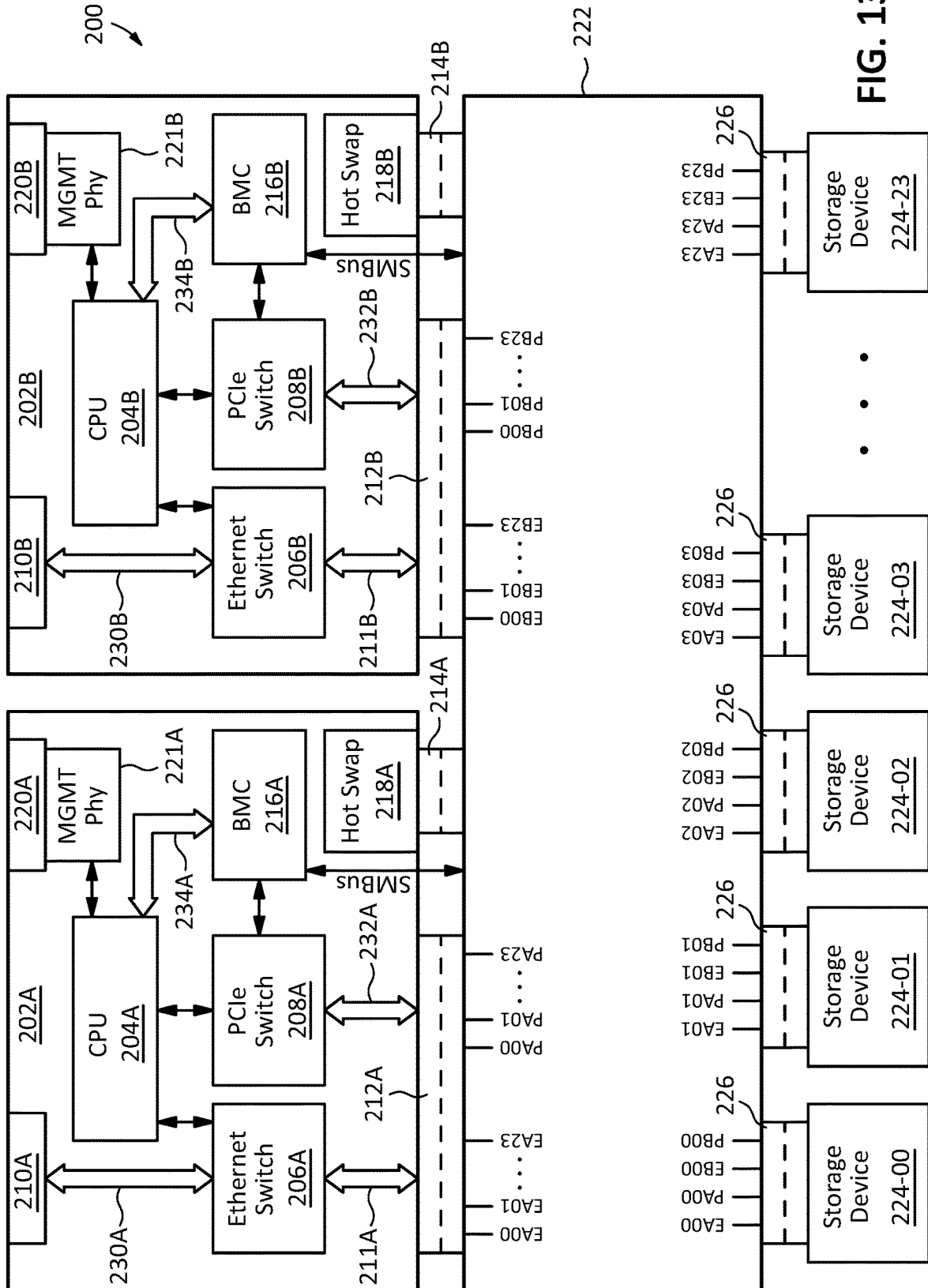


FIG. 12



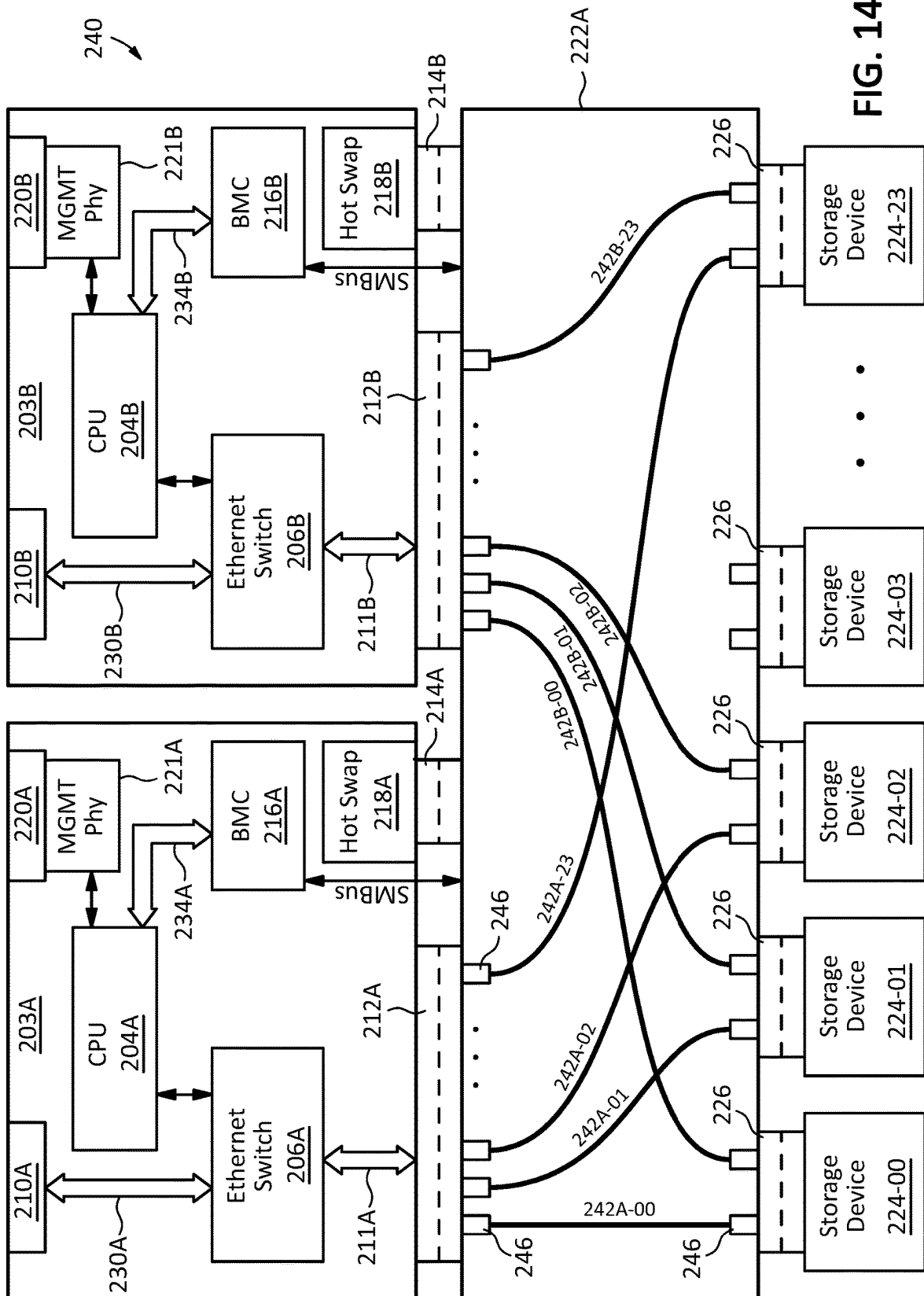
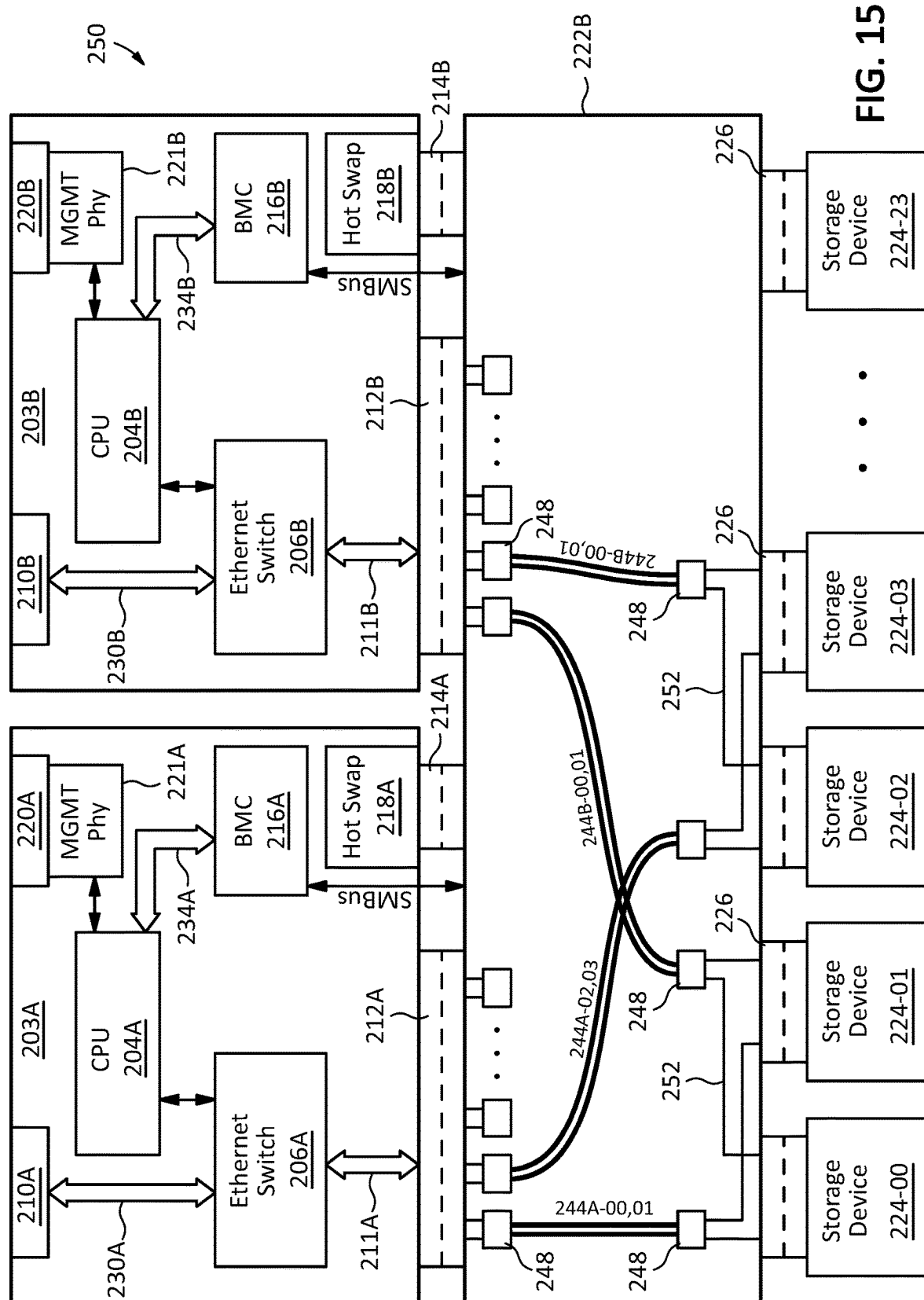


FIG. 14



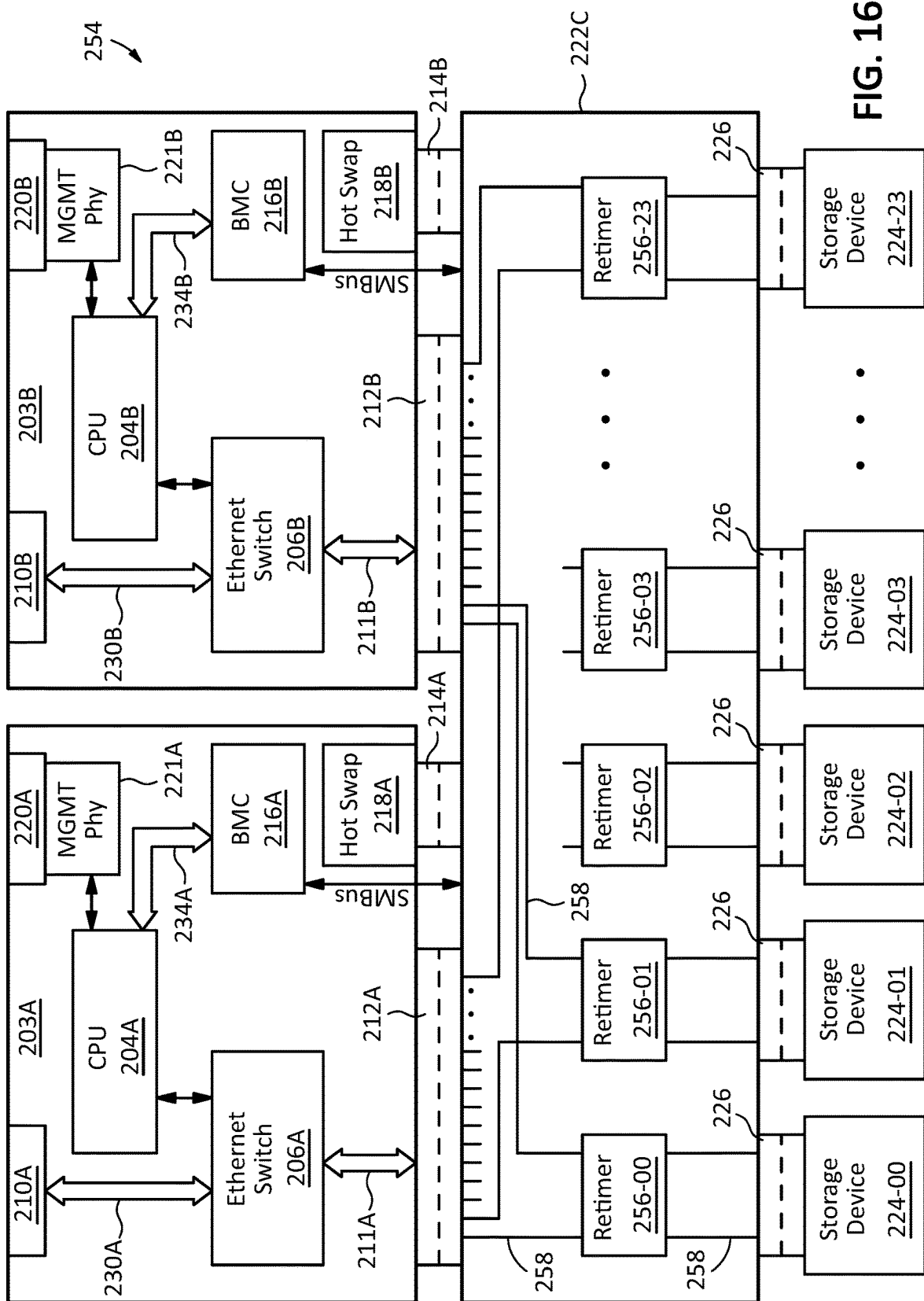


FIG. 16

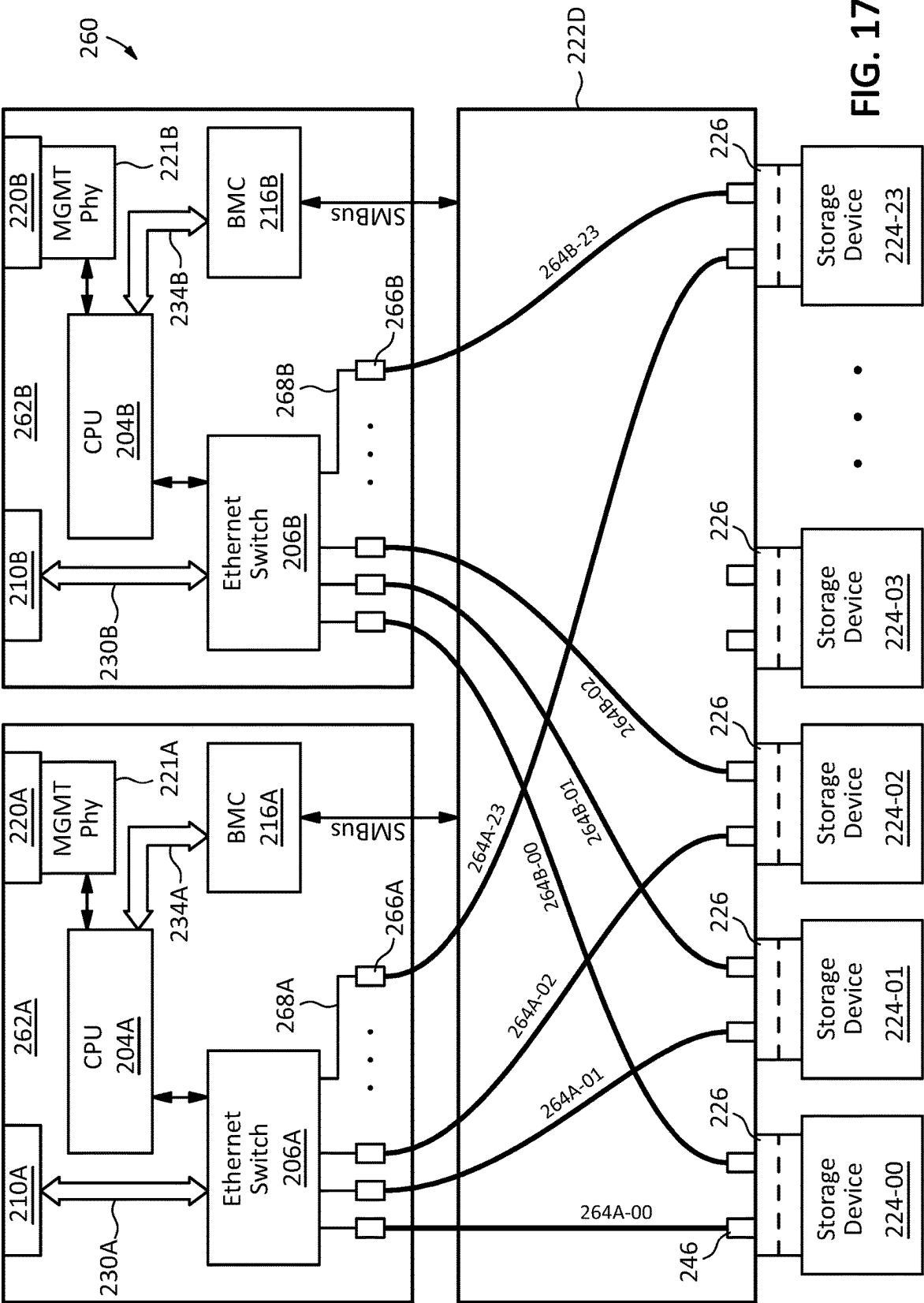
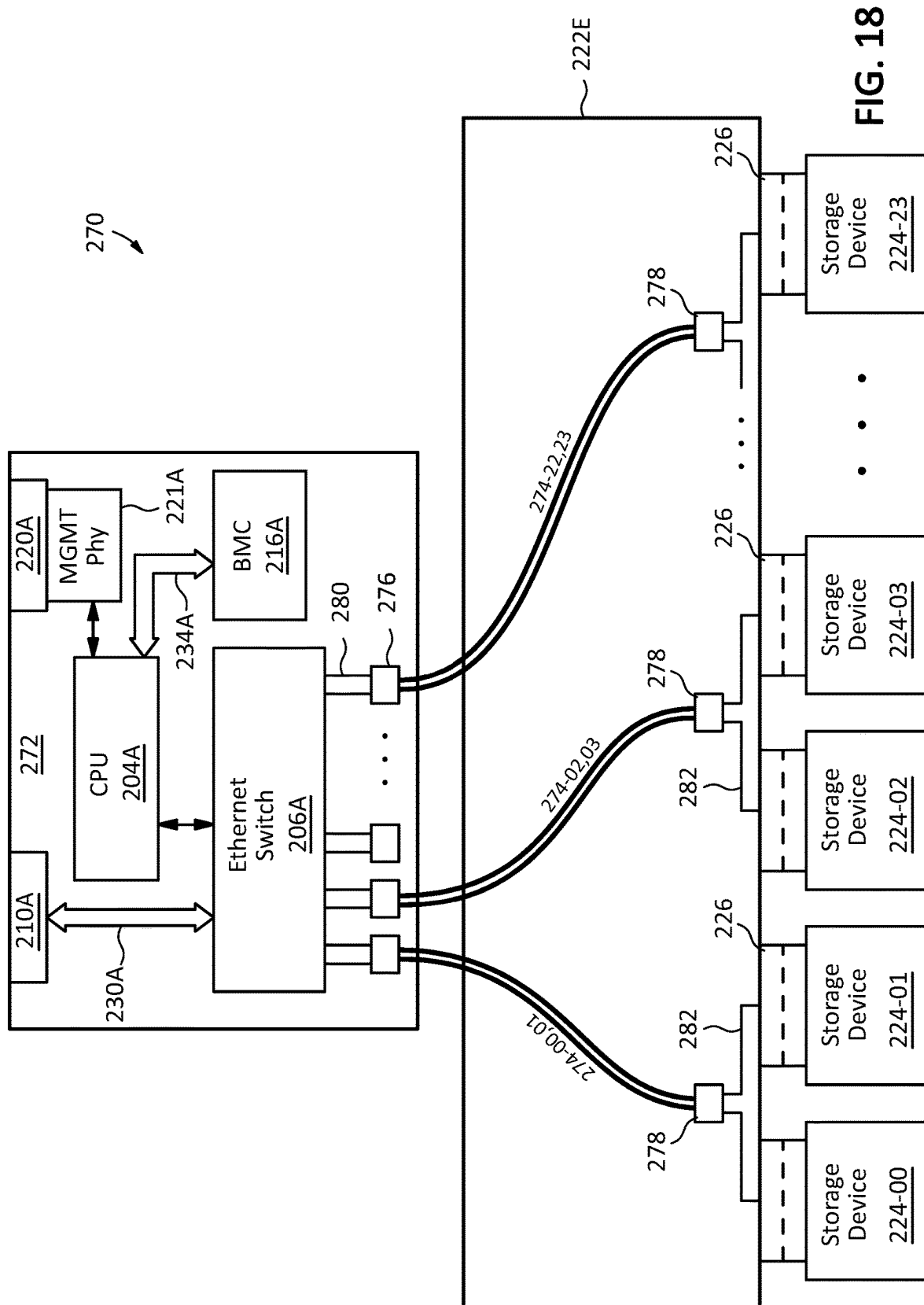


FIG. 17



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HIGH-SPEED DATA TRANSFERS THROUGH STORAGE DEVICE CONNECTORS

This application is a continuation of U.S. patent application Ser. No. 16/745,332, filed Jan. 16, 2020, which claims priority to, and the benefit of, U.S. Provisional Patent Application Ser. No. 62/821,429, filed Mar. 20, 2019, and U.S. Provisional Patent Application Ser. No. 62/924,689, filed Oct. 22, 2019, all of which are incorporated by reference.

BACKGROUND

1. Field

This disclosure relates generally to data storage systems, and more specifically to systems and methods for supporting data transfers through storage device connectors using high-speed protocols.

2. Related Art

Data storage devices such as hard disk drives (HDDs), solid state drives (SSDs), and the like, can be connected to storage systems through connectors that may enable the storage devices to be added to or removed from the system for expansion, maintenance, repair, upgrade, etc. One type of storage device connector known as a U.2 connector may be used in data storage systems, for example, in data centers and other large-scale deployments. U.2 connectors may enable a single connector design to be used with multiple storage protocols including Serial ATA (SATA), Serial Attached SCSI (SAS) and Non-Volatile Memory Express (NVMe).

SUMMARY

A storage device may include a multi-protocol storage device connector, and a storage controller configured to control data transfers to and from a storage medium through the multi-protocol storage device connector, wherein the storage controller may be coupled to the multi-protocol storage device connector through an interface for a high-speed multi-level signaling protocol. The multi-protocol storage device connector may be a U.2 connector. The U.2 connector may be a serial attached small computer system interface generation four (SAS4) capable U.2 connector. The high-speed multi-level signaling protocol may include an Ethernet protocol with pulse amplitude modulation level-4 (PAM4) modulation. The Ethernet protocol may be a 50 Gb Ethernet protocol. The high-speed multi-level signaling protocol may include a PCIe protocol such as PCIe Gen 6. The interface may be configured to implement a non-volatile memory express over fabric (NVMe-oF) protocol. The multi-protocol storage device connector may be compliant with the Small Form Factor 8639 (SFF-8639) specification.

A data storage system may include a switch board configured to communicate using a high-speed multi-level signaling protocol, and a midplane having one or more multi-protocol storage device connectors configured to couple the midplane to one or more storage devices, wherein the midplane may be coupled to the switch board and configured to enable the one or more storage devices to communicate with the switch board through the one or more multi-protocol storage device connectors using the high-speed multi-level signaling protocol. The midplane may be coupled to the switch board through one or more high-speed

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connectors. The midplane may include a circuit board having conductive traces configured to couple one or more of the high-speed connectors to one or more of the multi-protocol storage device connectors. The midplane may include one or more re-timers coupled between one or more of the high-speed connectors and one or more of the multi-protocol storage device connectors. One or more of the high-speed connectors and one or more of the multi-protocol storage device connectors may be coupled through one or more cables. The one or more cables may include one or more axial cables. One or more of the multi-protocol storage device connectors on the midplane may be coupled to the switch board through one or more cables. One or more of the cables may include a connector at one end configured to connect the cable to the switch board. One or more of the cables may include a connector at one end configured to connect the cable to the midplane.

A method for transferring data between a switch board and a storage device may include, transferring data between the switch board and a midplane using a high-speed multi-level signaling protocol, and transferring data between the midplane and the storage device using the high-speed multi-level signaling protocol, wherein the storage device may be coupled to the midplane through a multi-protocol storage device connector. The data may be transferred between the switch board and the midplane through one or more high-speed connectors. The method may further include re-timing signals for the high-speed multi-level signaling protocol on the midplane or the switch board. The data may be transferred between the switch board and the midplane through one or more conductive traces or cables.

A data storage midplane may include a circuit board, and one or more multi-protocol storage device connectors configured to couple the circuit board to one or more storage devices, wherein the midplane may be configured to enable data transfers between a switch board and a storage device through one of the multi-protocol storage device connectors using a high-speed multi-level signaling protocol. The midplane may further include one or more high-speed connectors configured to couple the midplane to the switch board and transfer data using the high-speed multi-level signaling protocol. The circuit board may include conductive traces configured to couple one or more of the high-speed connectors to one or more of the multi-protocol storage device connectors. The midplane or the switch board may include one or more re-timers coupled between one or more of the high-speed connectors and one or more of the multi-protocol storage device connectors. One or more of the high-speed connectors and one or more of the multi-protocol storage device connectors may be coupled through one or more cables. The one or more cables may include one or more axial cables. The midplane may be configured to couple one or more of the multi-protocol storage device connectors to the switch board through one or more cables. The circuit board may include one or more connectors for the one or more cables.

BRIEF DESCRIPTION OF THE DRAWINGS

The figures are not necessarily drawn to scale and elements of similar structures or functions are generally represented by like reference numerals for illustrative purposes throughout the figures. The figures are only intended to facilitate the description of the various embodiments described herein. The figures do not describe every aspect of the teachings disclosed herein and do not limit the scope of the claims. To prevent the drawing from becoming obscured,

not all of components, connections, and the like may be shown, and not all of the components may have reference numbers. However, patterns of component configurations may be readily apparent from the drawings. The accompanying drawings, together with the specification, illustrate example embodiments of the present disclosure, and, together with the description, serve to explain the principles of the present disclosure.

FIG. 1 illustrates an embodiment of a data storage system according to this disclosure.

FIG. 2 illustrates an embodiment of a data storage system in which a midplane may be connected to a switch board through one or more high-speed connectors according to this disclosure.

FIG. 3 illustrates an embodiment of a data storage system in which a midplane may use one or more cables according to this disclosure.

FIG. 4 illustrates an embodiment of a data storage system in which a midplane may use one or more re-timers according to this disclosure.

FIG. 5 illustrates an embodiment of a data storage system in which a midplane may be connected to a switch board through one or more cables according to this disclosure.

FIG. 6 illustrates an embodiment of a data storage system in which one or more re-timers may be located on a switch board according to this disclosure.

FIG. 7 illustrates an embodiment of a storage device according to this disclosure.

FIG. 8 illustrates another embodiment of a storage device according to this disclosure.

FIG. 9 illustrates another embodiment of a storage device according to this disclosure.

FIG. 10 illustrates a model of an embodiment of a communication channel for a data storage system according to this disclosure.

FIG. 11 illustrates insertion loss simulation results for an example embodiment of a communication channel according to this disclosure.

FIG. 12 illustrates return loss simulation results for an example embodiment of a communication channel according to this disclosure.

FIG. 13 illustrates an example embodiment of a data storage system illustrating some possible implementation details according to this disclosure.

FIG. 14 illustrates an example embodiment of a data storage system illustrating some possible implementation details including cables according to this disclosure.

FIG. 15 illustrates an example embodiment of a data storage system illustrating some possible implementation details including multi-channel cables according to this disclosure.

FIG. 16 illustrates an example embodiment of a data storage system illustrating some possible implementation details including re-timers according to this disclosure.

FIG. 17 illustrates an example embodiment of a data storage system illustrating some possible implementation details including cables according to this disclosure.

FIG. 18 illustrates an example embodiment of a data storage system illustrating some possible implementation details including multi-channel cables according to this disclosure.

DETAILED DESCRIPTION

U.2 connectors may be manufactured in accordance with the SFF-8639 specification which is published by the Small Form Factor (SFF) Committee of the Storage Networking

Industry Association (SNIA). The SFF-8639 specification defines the mechanical specifications and general requirements for U.2 plug and receptacle connectors for storage devices. A physical U.2 connection typically involves a U.2 plug on the storage device that may connect to a U.2 socket on a circuit board or cable. U.2 connectors can accommodate three families of storage protocols: SATA, SAS and NVMe, which may use Peripheral Component Interconnect Express (PCIe) as an underlying protocol, U.2 connectors may also support variants of these storage protocols.

U.2 connectors may also be used to transfer data to and from storage devices using network protocols such as Ethernet, Fibre Channel, InfiniBand, and others. For example, a storage device may be connected to a storage system through a U.2 connector and configured to transfer data using NVMe over Fabric (NVMe-oF) which is a specification that enables NVMe protocol to be transmitted over network fabrics such as Ethernet, Fibre Channel, InfiniBand, and others. This may be accomplished, for example, by using pins of the U.2 connector, which may normally be defined for use with a SAS or PCIe protocol, for implementing a channel for the network fabric.

The ability of the U.2 connector to accommodate these different protocols, for example, in data centers and other large-scale deployments, has resulted in adoption of the connector design and resulting mass production. This mass production, in turn, has provided economies of scale which have driven down the cost of U.2 connectors and made them available from different vendors.

Communication protocols may be revised periodically to provide higher data transfer rates with each generation of the protocol. As data transfer rates and accompanying signaling speeds increase, the connectors used for each protocol often need to be enhanced to accommodate the higher signal speeds. In some situations, the connectors may be enhanced through the use of different materials and/or manufacturing techniques and/or mechanical refinements without changing the overall mechanical design of the connectors. In other situations, it may not be possible or economically feasible to accommodate higher signal speeds without changing the overall mechanical design of the connectors.

As the speed of a communication protocol increases with each new generation, different techniques for encoding, signaling, modulation, sampling, receiver equalization, error correction, etc., may be adopted to enable higher data transfer rates. A group of signaling techniques known as multi-level signaling may increase the rate at which a communication protocol transfers data. Multi-level signaling may involve the use of more than two binary signal levels, e.g., signal levels between logic 0 and logic 1, to represent data. For example, four-level pulse amplitude modulation (PAM4) is a type of multi-level signaling that may use four different voltage levels to enable two bits of data to be represented by one symbol. This may effectively double the data transfer rate of a communication protocol, but it may result in significantly increased cost, complexity, and/or noise susceptibility. Therefore, multi-level signaling is typically not adopted until the challenges associated with multi-level signaling are outweighed by the need for higher data transfer rates. This may not occur, for example, until a protocol reaches a realm of high-speed operation in which data transfer rates may be specified in tens of gigabits per second. For example, PAM4 signaling was adopted in the transition from 25 Gb to 50 Gb Ethernet.

As the data transfer speeds of communication protocols increase, and multi-level signaling becomes more widespread, the U.2 connector may have reached a practical limit

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using conventional implementation techniques. However, with the wide availability and low cost of U.2 connectors, it may be beneficial to continue using U.2 connectors with high-speed communication protocols such as those that use multi-level signaling.

FIG. 1 illustrates an embodiment of a data storage system according to this disclosure. The system **100** shown in FIG. **1** may include a switch board **102**, a midplane **104**, and one or more storage devices **106**. Each of the storage devices **106** may be connected to the midplane **104** with a multi-protocol storage device connector **108**. The midplane **104** may be coupled to the switch board **102** and configured to enable the one or more storage devices **106** to communicate with the switch **102** board using a high-speed communication protocol **110** that may use multi-level signaling.

The switch board **102** may include any type of communication apparatus having at least one operating mode that uses a high-speed communication protocol with multi-level signaling. For example, in some embodiments, the switch board **102** may include an Ethernet switch capable of operating at 50 Gb speeds which may use pulse amplitude modulation level-4 (PAM4) signaling. In other embodiments, the switch board **102** may include a PCIe switch capable of operating at PCIe 6.0 speeds which may also use PAM4 signaling. In some additional embodiments, the switch board **102** may include a PCIe switch capable of operating at PCIe 4.0 or 5.0 speeds. In yet other embodiments, the switch board **102** may include combinations of these and/or other switches and/or communication apparatus.

In some embodiments, an Ethernet or other switch on switch board **102** may be a single node of a larger switch hierarchy or switch fabric. In other embodiments, the switch board **102** may include multiple switch nodes and/or interconnects for all or a portion of a switch fabric structure. Some embodiments may include additional switch boards. For example, multiple switch boards may be configured to provide each of the storage devices **106** with multiple network or interconnect ports.

The midplane **104** may include any type of circuit board **105** configured to couple the one or more storage devices **106** to the switch board **102**. This may include a standard midplane configuration in which the one or more storage device connectors **108** are arranged on one side of the board **105** and one or more other connectors or connection apparatus for coupling the midplane to the switch board **102** are arranged on the other side of the board **105**. As used herein, a midplane may also refer to any other type of circuit board configuration that may be used to physically and electrically couple the one or more storage devices **106** to the switch board **102**. This may include, for example, a backplane configuration in which both sets of connectors or connection apparatus are arranged on the same side of the board **105**. The midplane **104** may include a unitary circuit board **105** or it may be implemented with multiple boards and/or sets of connectors or connection apparatus.

The one or more storage devices **106** may include hard disk drives (HDDs), solid state drives (SSDs), and/or any other type of storage device based on magnetic, solid state, optical, or any other type of data storage technology. The one or more storage devices **106** may implement any type of storage interface and/or protocol having at least one operating mode that uses a high-speed protocol with multi-level signaling. For example, one or more of the storage devices **106** may be capable of transferring data through the connector **108** according to an NVMe over fabric (NVMe-oF) specification using 50 Gb Ethernet as the underlying fabric

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protocol which may use PAM4 signaling. As another example, one or more of the storage devices **106** may be capable of transferring data through the connector **108** according to an NVMe specification using PCIe 6.0 as the underlying protocol which may use PAM4 signaling.

The one or more storage device connectors **108** may include a multi-protocol storage device connector such as a U.2 connector. Although a connector **108** may be illustrated as a single component, it may typically include a plug or header portion, which may be included as part of a storage device **106**, and a receptacle portion, which may be included as part of the midplane **104**.

FIGS. 2-6 illustrate additional embodiments of data storage systems showing examples of techniques in which the physical and electrical coupling between a switch board **102** and midplane **104**, shown by arrow **103** in FIG. 1, may be implemented according to this disclosure.

FIG. 2 illustrates an embodiment of a data storage system **101** in which a midplane **104A** may be connected to a switch board **102A** through one or more high-speed connectors **112**. The one or more high-speed connectors **112** may include a mechanical design and selection of materials that may enable the connector to accommodate a high-speed communication protocol with multi-level signaling such as 50 Gb Ethernet. Although a high-speed connector **112** may be illustrated as a single component, it may typically include two different mating portions with one portion mounted on the switch board **102A** and the other portion mounted on the circuit board **105A** of the midplane **104A**.

The one or more high-speed connectors **112** may be electrically connected to the one or more storage device connectors **108** through traces, vias, and/or other conductive structures **114** on the circuit board **105A** which may form the foundation of the midplane **104A**. The traces, vias, and/or other conductive structures **114** which may be arranged, for example, as transmission lines such as slotlines, striplines, microstrips, and the like, on the circuit board **105A** that may be capable of carrying the high-speed communication protocol **110** that may use multi-level signaling.

FIG. 3 illustrates another embodiment of a data storage system **116** in which a midplane **104B** may be connected to a switch board **102A** through one or more high-speed connectors **112**. In the embodiment illustrated in FIG. 3, however, the one or more high-speed connectors **112** may be electrically connected to the one or more storage device connectors **108** at least partially through one or more cables **118**. The cables **118** may include coaxial cables, twin-axial cables, and/or any other cable-like conductive structures that may be capable of carrying the high-speed communication protocol **110** that may use multi-level signaling. One or more of the cables **118** may include cable-to-board connectors at either or both ends which may be snapped, soldered, crimped or otherwise connected to the circuit board **105B** to make electrical and/or mechanical connections. Alternatively, or additionally, one or more of the cables **118** may be connected at one or both ends to one or more of the high-speed connectors **112** and/or storage device connectors **108**. One or more of the cables **118** may include cable-to-connector connectors at either or both ends which may be snapped, soldered, crimped or otherwise connected to any of the high-speed connectors **112** and/or storage device connectors **108** to make electrical and/or mechanical connections.

FIG. 4 illustrates another embodiment of a data storage system **120** in which a midplane **1040** may be connected to a switch board **102A** through one or more high-speed connectors **112**. In the embodiment illustrated in FIG. 4,

however, the midplane **1040** may include one or more re-timers **122** connected between one or more of the high-speed connectors **112** and one or more of the storage device connectors **108**. The re-timers **122** may be capable of re-timing the high-speed communication protocol with multi-level signaling **110** in either or both directions. The re-timers **122** may be connected to one or more of the high-speed connectors **112** through conductors **114A**. The re-timers **122** may be connected to one or more of the storage device connectors **108** through conductors **114B**. The conductors **114A** and **114B** may be implemented as traces, vias, and/or other conductive structures which may be arranged, for example, as transmission lines such as slotlines, striplines, microstrips, and the like, on the circuit board **1050** that may be capable of carrying the high-speed communication protocol **110** that may use multi-level signaling. Alternatively, or additionally, one or more of the re-timers **122** may be connected to one or more of the high-speed connectors **112** and/or one or more of the storage device connectors **108** at least partially through one or more cables as described above with respect to the system **116** illustrated in FIG. 3.

FIG. 5 illustrates an embodiment of a data storage system **124** in which a midplane **104D** may be connected to a switch board **102B** at least partially through one or more cables **126**. The cables **126** may include coaxial cables, twin-axial cables, and/or any other cable-like conductive structures that may be capable of carrying the high-speed communication protocol **110** that may use multi-level signaling. One or more of the cables **126** may include cable-to-board connectors at either or both ends which may be snapped, soldered, crimped or otherwise connected to the circuit board **105D** and/or the switch board **102E** to make electrical and/or mechanical connections. Alternatively, or additionally, one or more of the cables **126** may be connected at one or both ends to one or more of the storage device connectors **108** or one or more headers or other connectors such as a network or storage interface connectors on the switch board **102B**. One or more of the cables **126** may include cable-to-connector connectors at either or both ends which may be snapped, soldered, crimped or otherwise connected to any of storage device connectors **126** and or other connectors such as a network or storage interface connectors on the switch board **102B** to make electrical and/or mechanical connections.

The one or more cables **126** may be combined one or more connectors to create hybrid configurations of cables and connectors to couple the midplane **104D** to the switch board **102B**. For example, one or more connectors may be used to provide power and/or lower speed connections and/or a mechanical connection between the midplane **104D** and the switch board **102B**, while the one or more cables **126** may provide high-speed signal connections.

FIG. 6 illustrates another embodiment of a data storage system **125** in which one or more re-timers **122A** may be located on the switch board **1020** instead of, or in addition to, the midplane **104E**. Locating re-timers **122A** on the switch board **1020** may be beneficial, for example, in implementations in which the switch board may be easier to replace than the midplane. Thus, active circuitry in the re-timers **122A** may be easier to replace. The physical and electrical coupling between the switch board **1020** and midplane **104E**, shown by arrow **123**, may be implemented with any suitable arrangement of components such as, for example, any of the connectors, conductive traces, cables, and the like, described in the context of the embodiments above and below according to this disclosure.

FIG. 7 illustrates an embodiment of a storage device **128** according to this disclosure. The storage device **128** may be used, for example, as one of the storage devices **106** illustrated in FIGS. 1-6. The storage device **128** may include a storage device connector **130**, a communication interface **132**, a storage controller **134** and storage media **136**. The storage device connector **130** may include a multi-protocol storage device connector such as a U.2 connector. For example, the connector **130** may be implemented as a U.2 plug which may enable the storage device **128** to be plugged into a corresponding U.2 receptacle on a midplane or other apparatus in a data storage system. The communication interface **132** may provide access to the storage controller **134** through the storage device connector **130**. The communication interface **132** may implement a high-speed communication protocol that uses multi-level signaling such as 50 Gb Ethernet or PCIe 6.0. The storage media **136** may be implemented with magnetic, solid state, optical, or any other type of data storage technology. The communication interface **132** and a storage controller **134** may be implemented as separate components or integrated into a single component. Either of the communication interface **132** and storage controller **134** may be implemented in hardware, software or any combination thereof.

FIG. 8 is a block diagram of another embodiment of a storage device **138** illustrating some possible implementation details according to this disclosure. In this example embodiment, the storage device **138** may be implemented as a solid state drive (SSD) using an array of flash memory devices **140** as the storage media, and a U.2 connector **142** as the storage device connector **142**. A storage controller **144** may include a flash translation layer (FTL) **146** as well as an NVMe manager **148**. A communication interface **150** may be implemented, for example, with a field programmable gate array (FPGA) or application specific integrated circuit (ASIC) configured to implement one or more 50 Gb Ethernet ports **152** accessible through the U.2 connector **142**. The communication interface **150** may enable the NVMe storage protocol from the storage controller **144** to be transported through the one or more 50 Gb Ethernet ports **152**, thereby implementing NVMe-oF. In this configuration, pins of the U.2 connector which may normally be defined for use with a SAS or PCIe protocol may be used as a data plane for the one or more Ethernet ports **152**, while unused pins normally defined for use as PCIe lanes may be used for a control plane for the one or more Ethernet ports **152**.

FIG. 9 is a block diagram of another embodiment of a storage device **138** illustrating some additional possible implementation details according to this disclosure. The device **138** illustrated in FIG. 9 may be similar to the embodiment illustrated in FIG. 8 but with the addition of another communication interface **156** which may implement another communication protocol. For example, the second interface **156** may implement one or more PCIe ports **158**, while the first interface **150** implements one or more 50 Gb Ethernet ports **152**. Thus, the storage device may be implemented as a multi-mode device capable of operating in different modes using different communication protocols. For example, if the storage device **138** is implemented as a solid state drive, the two communication interfaces may enable it to operate as either an NVMe device or an NVMe-oF device while using the same U.2 connector in either mode. Moreover, the second interface **156** may implement a PCIe protocol that may also use multi-level signaling such as PAM4 in PCIe Gen 6. Thus, in some embodiments, the storage device **138** may be able to operate as either an NVMe device or an NVMe-oF device, both of which use a

multi-level protocol such as PAM4. In other embodiments, the interfaces **150** and **156** may implement any other suitable combination of communication protocols. For example, either or both interfaces may implement any a network protocol such as Ethernet, Fibre Channel, InfiniBand, and the like, and either or both interfaces may implement a storage protocol such as PCIe, SAS, DATA, and the like, wherein one of the protocols may be a high-speed multi-level signaling protocol.

Multi-mode operation may be implemented, for example, by using a vendor-defined pin or reserved future use (RFU) pin as a chassis-type or protocol-selector pin. The state of the vendor defined or RFU pin on a U.2 connector, for example, may be hard-wired into the board on which the connector is mounted. As another example, a mode control pin may be controlled by logic on or connected to the board. As yet other examples, the mode of the storage drive **138** may be controlled by an in-band command, through a management control bus, or any other technique for controlling the operating mode.

In some embodiments, the storage device **138** may operate in an NVMe mode using one or more of the PCIe ports connected through pins defined for PCIe signals on the U.2 connector. The storage device **138** may operate in an NVMe-of mode using pins normally defined for SAS signals to carry Ethernet or other network signals as a data plane and pins normally defined for PCIe lanes for a control plane. For example, a central processor unit (CPU), a baseboard management controller (BMC), and/or other devices may use PCIe lanes as a control plane to communicate with the storage device **138**.

In some embodiments, the storage device **138** may use a U.3 connector, which may have the same physical design as a U.2 connector, but which may be used with a specification that may provide different pin definitions. In some embodiments, the storage device **138** may use a U.3 connector, but with pin definitions for a U.2 connector.

Any of the components and/or functionality described in the embodiments illustrated in FIGS. **8** and **9** may be implemented with hardware, software, or any combination thereof. Moreover, any of the various components may be implemented as separate components, integrated together, or arranged in any other suitable configuration. In some embodiments, if the communication interfaces are implemented with an FPGA or other logic device, the storage device may be fabricated conveniently and/or economically by adding the logic device to an existing storage device, thereby potentially reducing the manufacturing cost, time-to-market, and/or total cost of ownership (TCO) of the storage device.

Depending on the implementation details, the storage devices according to this disclosure may provide various additional benefits and advantages. For example, a storage device may be versatile because it may be used as either an NVMe or NVMe-of device, with the further potential advantage of being able to operate with a high-speed communication protocol that may use multi-level signaling such as 50 Gb Ethernet and/or PCIe 6.0. The principles of this disclosure may also reduce the per unit cost and/or TCO, for example, because of economies of scale resulting from being able to use the same storage device as an NVMe or NVMe-of device. The principles of this disclosure may also reduce the time-to-market of a storage device and/or storage system. Moreover, some or all of the features may be implemented using functionality that may already exist inside a system chassis and thus may have reduced cost in comparison with other approaches.

FIG. **10** illustrates a model **160** of an embodiment of a communication channel **160** for a data storage system according to this disclosure. The model illustrated in FIG. **10** is described in the context of some specific types and arrangement of components, protocols, and the like, but the principles of this disclosure are not limited to these details. Not every component illustrated in FIG. **10** may be needed for purposes of modeling and other embodiments may omit some components and/or include additional components. The model illustrated in FIG. **10** may be used, for example, to perform a signal integrity (SI) analysis of a communication channel using the high-speed protocol **110** with multi-level signaling in the system illustrated in FIG. **2**.

Referring to FIG. **10**, the model of the channel **160** may include a model of a differential output driver **162** on an integrated circuit (IC) mounted on a switch board. The model may also include transmission lines **Z1A** and **Z1B** and capacitors **C1A** and **C1B** which may couple the differential outputs of the output driver **162** to vias **168A** and **168B**. Although referred to as vias, the structures **168A** and **168B**, as well as other vias mentioned below, may also be realized as plated through holes, solder pads, contact pads, or any other type of structure that may provide an electrical connection between conductive traces on layers of a circuit board and/or between conductive traces and terminals of components such as ICs, resistors, capacitors, connectors, sockets, etc.

The model **160** may further include transmission lines **Z2A** and **Z2B** which may model circuit board traces between the vias **168A** and **168B** and vias **172A** and **172B**, respectively. The output driver **162** and transmission lines **164A**, **1648**, **170A** and **170E** may form all or a portion of a switch fabric **161** on the switch board. The vias **172A** and **1728** may be used for mounting a receptacle portion **1748** of a high-speed connector to the switch board. A plug or header portion **174P** of the high-speed connector may be mounted to the circuit board of a midplane **178** using vias **176A** and **1768**. The receptacle portion **1748** and plug or header portion **174P** of the high-speed connector may be referred to collectively as **174**. Transmission lines **Z3A** and **Z3B** may model conductive traces on the midplane **178** to vias **182A** and **1828**, which may be used to mount a receptacle portion **1848** of a storage device connector to the midplane **178**. A plug portion **184P** of the storage device connector may be mounted to a circuit board inside a storage device **186** using vias **188A** and **1888**. The receptacle portion **184R** and plug portion **184P** of the storage device connector may be referred to collectively as **184**. Transmission lines **Z4A** and **Z4B** may model conductive traces between the vias **188A** and **188E** and vias **192A** and **1928**, respectively, which may be connected to the differential inputs of a receiver **194** on an IC in the storage device **186**.

The model **160** illustrated in FIG. **10** may be replicated in the other direction for data flowing from the storage device **186** to the switch fabric **161**.

The model **160** illustrated in FIG. **10** may be used for an SI simulation of an example embodiment of a communication channel which may be implemented with some or all of the following details. These details are provided for purposes of illustration, but the principles of this disclosure are not limited to these details.

In this example embodiment, the output driver **162** may be part of an IC having a 50 Gb Ethernet switch driving the switch fabric **161**. By way of example, the Ethernet switch may comply with Institute of Electrical and Electronics Engineers (IEEE) standard 802.3cd for 50 Gb Ethernet. The Ethernet switch may further comply with the IEEE 802.3cd

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physical layer specification 50 GBASE-KR for transmission over a single lane of an electrical backplane. The physical layer may operate at approximately 56 Gb/s with PAM4 modulation. For example, an approximately 14 GHz signal may use two data eyes per clock cycle to provide a 28 GHz baud rate, i.e., symbols per second. By using four-level signaling, the PAM4 modulation may enable the transfer of two bits per symbol, thereby providing a data rate of about 56 Gb/s.

In this example embodiment, the high-speed connector **174** may be fabricated with conductors made from various high-performance alloys of copper and/or using gold plating on contacts. It may include a connector body made from high-performance dielectric materials and designed with a mechanical arrangement that may avoid placement of dielectric materials in areas of concentrated electromagnetic fields that may slow down and/or interfere with signal propagation. Examples of additional possible characteristics of a high-speed connector include the following: linear transmission beyond 40 GHz, differential pairs of terminals, mechanically and/or electrically matched signal lengths within differential pairs of terminals, wafer-based design, high terminal density, reduced or no stub resonances from signal contacts, and/or damping shielding, for example, 3D resonance damping shielding that may enable low crosstalk across a wide frequency spectrum. In some implementations, a high-speed connector may be specified for operation at about 112 Gbs.

In this example embodiment, the storage device connector **184** may be implemented with a U.2 connector, for example, a SAS4 capable U.2 connector. A SAS4 capable U.2 connector may be made with materials and construction techniques that may enable it to operate at clock speeds specified by the fourth generation SAS protocol and/or later generations of SAS protocols or similar protocols.

In some implementations of this example embodiment, the lengths of some or all of the circuit board traces may be kept below certain limits. For example, the traces on the switch board modeled by transmission lines **170A** and **170B** may be limited to about four inches in length. As another example, the traces on the midplane **178** modeled by transmission lines **188A** and **188B** may be limited to about 13 inches in length.

In this example embodiment, the receiver **194** may be part of an FPGA in the storage device **186** that may implement an interface for 50 Gb Ethernet. For example, if the storage device **186** is implemented as an SSD, the FPGA may enable the SSD to transfer data through the connector **186** according to the NVMe-oF specification using 50 Gb Ethernet as the underlying protocol.

FIGS. **11** and **12** illustrate examples of the type of channel performance simulation results that may be obtained using the model of the channel **160**. The results shown in FIGS. **11** and **12** are for purposes of illustrating the principles of this disclosure and may not represent the performance results of a specific physical or simulated embodiment.

FIG. **11** illustrates an insertion loss (IL) mask, i.e., limit of allowable insertion loss, for an exemplary channel specification for 50 Gb Ethernet and simulation results for an implementation of an example embodiment using the model illustrated in FIG. **10**. FIG. **12** illustrates a return loss (RL) mask, i.e., limit of allowable return loss, for the exemplary channel specification and simulation results for the same implementation as used in FIG. **11**. In both figures, the channel specification is shown as a dashed line, and the simulation results are shown as a solid line. As may be apparent from FIGS. **11** and **12**, the simulated implementa-

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tion may comply with the IL and RL specifications at the relevant operating frequencies. Moreover, the simulated implementation may also comply with an effective return loss (ERL) specification which may include a software component.

FIG. **13** illustrates an example embodiment of a data storage system **200** illustrating some possible implementation details according to this disclosure. The system **200** may include two switch boards **202A** and **202B**. A first one of the switch boards **202A** may include a CPU **204A**, an Ethernet switch **206A**, a high capacity or speed Ethernet connector **210A** for an uplink such as a QSFP28 connector, one or more high-speed connectors **212A**, one or more additional connectors **214A**, a baseboard management controller (BMC) **216A**, a hot swap controller **218A**, a system management connector **220A**, and a remote system management physical interface **221A**.

The first switch board **202A** may be connected to a midplane **222** through the one or more high-speed connectors **212A** and the one or more additional connectors **214A**. The second switch board **202B** may be connected to the midplane **222** through the one or more high-speed connectors **212B** and the one or more additional connectors **214B**. The high-speed connectors **212A** and **212E** may be implemented, for example, using connectors having some or all of the characteristics of the connector **174** described above with respect to FIG. **10** which may enable the high-speed connectors **212A** and **212B** to be used with high-speed communication protocols that may use multi-level signaling.

Storage devices **224-00** through **224-23**, which may be referred to collectively as **224**, may be connected to the midplane **222** through storage device connectors **226**. In this example, there may be up to 24 or more storage devices, but the system may be configured to accommodate any number of storage devices. Although this configuration is not limited to any specific type of storage devices or connectors, in this example embodiment, connectors **226** may be SAS4 capable U.2 connectors, and the storage devices may be SSDs having FPGAs configured to provide interfaces for both 50 Gb Ethernet and PCIe (e.g., for NVMe data transfers as described above with respect to the embodiment of FIG. **9**).

On the first switch board **202A**, the Ethernet switch **206A** may be connected to the one or more high-speed connectors **212A** through conductive traces, cables, or any suitable connection arrangement **211A** that may provide a 50 Gb Ethernet lane for each of the storage devices **224**, for example 24×50 GBASE-KR. On the upstream side, the Ethernet switch **206A** may be connected to the high capacity or speed Ethernet connector **210A** through traces, cables, or any suitable connection arrangement **230A** to support network traffic for the downstream storage devices **224**. The high capacity Ethernet connector **210A** may include, for example, multiple QSFP28 connectors.

The PCIe switch **208A** may be connected to the one or more high-speed connectors **212A** through conductive traces, cables, or any suitable connection arrangement **232A** to provide, for example, four lanes of PCIe for each of the storage devices **224**. PCIe clock distribution functionality may be provided separate from, or integral with, the PCIe switch **208A**. In some embodiments, one or more of the storage devices **224** may be capable of multi-mode operation. For example, in a mode such as NVMe-oF which may use a networking protocol, some or all PCIe lanes and/or pins of one or more of the connectors **212A/B** and/or **216** may be used for the network protocol. In such a mode, any unused PCIe lanes and/or pins may be used for other purposes, for example, for a control plane.

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The CPU **204A** may include local supporting resources such as memory, local storage, a power source, etc. The CPU **204A** may perform various system management functions, as well as enabling data transfers to the storage devices **224** through the PCIe switch **208A**, for example using an NVMe protocol. The CPU **204A** may be connected to the Ethernet switch **206A**, PCIe switch **208A**, BMC **216A**, and system management physical interface **221A** through links having various numbers of PCIe lanes of various generational speeds as may be appropriate for the expected traffic as shown by the arrows in FIG. **13**. The BMC **216A** may also be connected to the PCIe switch **208A** through a PCIe link as shown by the arrow in the figure. The BMC **216A** may be connected to the CPU **204A** through a low pin count (LPC) interface **234A**.

A system management bus (SMBus) may be connected between the BMC **216A** and the midplane **222** and/or storage devices **224** through one of the additional connectors **214A**, through a dedicated connector, or in any other suitable manner. The one or more additional connectors **214A** may provide a board-to-board power connection between the hot swap controller **218A** and the storage devices **224** through the midplane **222**.

The second switch board **202B** may include essentially the same components and connections as the first switch board **202A**, but with reference numbers ending in B instead of A.

On the midplane **222**, the one or more high-speed connectors **212A** and **212B** may be connected to any or all of the U.2 connectors through conductive traces on the circuit board of the midplane. Channel connections between two connectors may be indicated by corresponding reference designators. For example, channel EA00 from the Ethernet switch **206A** on the first switch board **202A** may be connected to the first storage device **224-00** on the left side of FIG. **13** and so on. Likewise, channel PB00 from the PCIe switch **208B** on the second switch board **202B** may be connected to the first storage device **224-00** on the left side of FIG. **13** and so on.

In this example embodiment, the midplane traces may be routed so that a 50 Gb Ethernet lane using, for example PAM4 50 GBASE-KR, may connect each of the Ethernet switches **206A** and **206B** with each of the storage devices **224**. Thus, each of the storage devices **224** may have a dual port 503g Ethernet connection through its U.2 connector. The midplane traces may also be routed so that a four-lane PCIe link connects each of the PCIe switches **208A** and **208B** with each of the storage devices **224**. Thus, each of the storage devices **224** may have a dual port PCIe connection through its U.2 connector. Moreover, the use of high-speed connectors **212A** and **212B** and SAS4 capable U.2 connectors **226** may also enable the PCIe connections to operate at Gen 6 speeds which may use PAM4 multi-level signaling.

The system **200** illustrated in FIG. **13** may also enable the storage devices **224** to operate as multi-mode devices that may support either NVMe or NVMe-oF data transfer protocols by detecting information from a known location such as a chassis-type pin or protocol-selector pin. For example, a storage device **224** may operate in either an NVMe mode using one or more of the PCIe links or in an NVMe-oF mode using one or more of the Ethernet ports in response to the state of a vendor defined or a reserved future use (RFU) pin on the U.2 connector which may be used as a chassis-type or protocol-selector. The state of the vendor defined or RFU pin on a U.2 connector, for example, may be hard-wired into the midplane **222** and/or controlled by the SMBus.

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The system **200** illustrated in FIG. **13** is not limited to the specific implementation details described above. For example, although the system is shown with two switch boards **202A** and **202B**, the system may be implemented with only one or with more than two switch boards. As a further example, although 50 Gb Ethernet and PCIe 6.0 are described as possible implementation details, the system **200** may also enable the use of other high-speed communication protocols, including high-speed protocols that may use multi-level signaling. As yet another example, either the Ethernet switch **206A** or the PCIe switch **208A** may be omitted from the first switch board **202A**. Likewise, either the Ethernet switch **206B** or the PCIe switch **208B** may be omitted from the second switch board **202B**.

FIG. **14** illustrates another example embodiment of a data storage system **240** illustrating some possible implementation details according to this disclosure. The system **240** may be similar to the system **200** illustrated in FIG. **13**, but in the system illustrated in FIG. **14**, cables may be used instead of, or in addition to, at least some of the conductive traces on the midplane **222A** to carry signals for the high-speed communication protocols with multi-level signaling. In the embodiment illustrated in FIG. **14**, the switch boards **203A** and **203B** are shown without PCIe switches which may simplify the connections on the backplane **222A**, but in other implementations, PCIe switches may be included in addition to, or in place of, the Ethernet switches **206A** and **206B**.

The system **240** may include cables **242A-00** through **242A-23**, each of which may provide a signal path for an Ethernet channel between the one or more high speed connectors **212A** and one of the U.2 connectors **226**. The system **240** may also include cables **242B-00** through **242B-23**, each of which may provide a signal path for an Ethernet channel between the one or more high speed connectors **212E** and one of the U.2 connectors **226**. The cables may be implemented with coaxial cables, twin-axial cables, or any other cable-like conductive structures that may be capable of carrying 50 Gb Ethernet or other high-speed communication protocol that may use multi-level signaling. The cables are shown having connectors **246** at either end of the cable which may form a removeable or non-removeable mechanical and/or electrical connection that may be secured with snap-in-place construction, twist-locks, mating connectors or headers, solder, screws, or any other attachment technique. The connectors **246** may perform any or all of numerous additional functions including providing strain relief, channel identification, controlled termination of the cables including impedance matching, reducing reflections, etc. To prevent the drawing from becoming obscured, not all of the cables and other components are shown, and not all of the components have reference numbers. However, the patterns of component configurations are readily apparent from the drawing.

In some embodiments, the connectors **246** may be attached to the circuit board of the backplane **222A** and conductive traces may be used to complete the connections between the cables and the one or more high-speed connectors **212A** and/or U.2 connectors **226**. In some other embodiments, the connectors **246** may be configured to connect directly to some or all of the one or more high-speed connectors **212A** and/or U.2 connectors **226**.

In some embodiments, either or both ends of the cables may be connected directly to the circuit board of the backplane **222A** and/or the one or more high-speed connectors **212A** and/or U.2 connectors **226** without the connectors **246**. The cables may be attached directly by soldering, crimping or any other suitable technique.

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Some embodiments may include hybrid arrangements of cables and/or traces to connect the one or more high speed connectors **212A** and U.2 connectors **226**. For example, conductive traces may be used for shorter connections between connectors that may be relatively close on the midplane **222A**, while cables may be used for connections between connectors that are further apart and thus would require longer traces. Depending on the implementation details, the use of cables may improve rate at which data may be transmitted between the switch board and the storage devices by, for example, reducing insertion losses, return losses and other losses associated with transmission lines, noise, cross-talk, reflections, and the like.

FIG. **15** illustrates another example embodiment of a data storage system **250** illustrating some possible implementation details according to this disclosure. The system **250** may be similar to the system **240** illustrated in FIG. **14**, but the cables may be implemented with multi-channel cable assemblies that may include one coaxial, twin-axial, or other type of cable for each of the multiple channels. In the example embodiment illustrated in FIG. **15**, each assembly may include two cables, each of which may have a multi-channel connector **248** at either end. The connectors **248** may be connected to the midplane **222B** with headers, sockets, patterns of plated through holes, mating connectors, or with any other suitable technique. The multi-channel cable connectors **248** may be connected to the high-speed connectors **212A** and **212B** and/or storage device connectors **226** with conductive traces **252** on the midplane **222B**.

The system **250** may include multi-channel cables **244A-00,01** through **244A-22,23**, each of which may provide a signal paths for a pair of Ethernet channels between the one or more high speed connectors **212A** and one of the U.2 connectors **226**. The system **250** may also include multi-channel cables **244B-00,01** through **244B-22,23**, each of which may provide a signal paths for a pair of Ethernet channels between the one or more high speed connectors **212B** and one of the U.2 connectors **226**. To prevent the drawing from becoming obscured, not all of the cables and other components are shown, and not all of the components have reference numbers. However, the patterns of component configurations are readily apparent from the drawing.

As with the other embodiments described above, the embodiment illustrated in FIG. **15** may be implemented with a wide range of variations. For example, cable assemblies with 4, 8, 16 or any other number of cables per assembly may be used. In some embodiments, either or both ends of the cables in the multi-cable assemblies may be connected directly to the circuit board of the backplane **222B** and/or the one or more high-speed connectors **212A** and **212B** and/or U.2 connectors **226** without the connectors **248**. The cables may be attached directly by soldering, crimping or any other suitable technique. In some other embodiments, the connectors **248** may be configured to connect directly to some or all of the one or more high-speed connectors **212A** and/or U.2 connectors **226**.

FIG. **16** illustrates another example embodiment of a data storage system **254** illustrating some possible implementation details according to this disclosure. The system **254** may be similar to the systems illustrated in FIGS. **13-15**, but may include one or more re-timers **256-00** through **256-23** on the midplane **222C**. The re-timers may implement any number of the following functions: equalization such as adaptive equalization and decision-feedback equalization (DFE), either of which may be linear and/or programmable; clock data recovery (CDR); and offset calibration. The re-timers may compensate for transmission line losses and inter-

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symbol interference. They may also compensate for deterministic and/or random jitter. Any of these functions may be implemented in receiver and/or transmitter paths.

Each of the re-timers **256-00** through **256-23** illustrated in FIG. **16** may be arranged between one of the storage devices **224** and both of the switch boards to provide dual-port Ethernet to each of the storage devices **224**. This may be implemented, for example, with four-lane re-timer ICs which may use two lanes for each port, i.e., one lane in each direction for each port. The re-timers may be connected to the high-speed connectors **212A** and **212B** and/or the storage device connectors **226** through conductive traces **258** on the midplane. In other embodiments, different numbers of re-timers having different numbers of lanes may be used.

Depending on the implementation details, the use of re-timers may improve the rate at which data may be transmitted between the switch board and the storage devices by, for example, by transmitting cleaner signals downstream from the re-timers. Some embodiments may include hybrid arrangements of re-timers and/or traces to connect the one or more high speed connectors **212A** and **212B** with U2 connectors **226**. For example, conductive traces alone may be used for shorter connections between connectors that may be relatively close on the midplane **222C**, while re-timers may be used for connections between connectors that are further apart and thus may require longer traces. To prevent the drawing from becoming obscured, not all of the re-timers and/or other components are shown, and not all of the components have reference numbers. However, the patterns of component configurations are readily apparent from the drawing.

FIG. **17** illustrates another example embodiment of a data storage system **260** illustrating some possible implementation details according to this disclosure. The system **260** may be similar to the systems illustrated in FIGS. **14** and **15**, but in this embodiment, cables may directly connect storage device connectors **226** on the midplane **222D** to one or both of the switch boards **262A** and **262B**. In the embodiment illustrated in FIG. **17**, each of cables **264A-00** through **264A-23** may be connected between the first switch board **262A** and one of the storage device connectors **226**. Each of cables **264B-00** through **264B-23** may be connected between the second switch board **262B** and one of the storage device connectors **226**. The cables may have a connector **226A** or **226B** on the switch board side, and a connector **246** on the midplane side. The connectors **226A** and **226B** may be connected to the corresponding Ethernet switches **206A** and **206B** through traces **268A** and **268B**, respectively. The connectors **246** may be connected directly to the storage device connectors **226**.

As with the other embodiments described above, the cables may be connected at either end, with or without the use of connectors and/or conductive traces, using any suitable connection technique. However, because the cables may be connected between the switch boards and midplane without the high-speed connectors used in other embodiments, it may be beneficial to use a cable configuration that may be easily disconnected and reconnected at one or both ends. This may include, for example, the use of cable termination connectors that snap in to mating connectors and/or patterns of conductors on the circuit boards.

Depending on the implementation details, connecting the cables directly to one or both of the switch boards may improve rate at which data may be transmitted between the switch boards and the storage devices by, for example, reducing resistive losses, insertion losses, return losses and other losses as well as noise, cross-talk, reflections, and the

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like that may be associated with the use of connectors between the switch boards and the midplane,

FIG. 18 illustrates another example embodiment of a data storage system 270 illustrating some possible implementation details according to this disclosure. The system 270 may be similar to the system illustrated in FIG. 17, but in this embodiment, multi-channel cable assemblies 274-00,01 through 274-22,23 may be used to connect pairs of storage device connectors 262 directly to a switch board 272. The cables may have a connector 276 on the switch board side, and a connector 278 on the midplane side. The connectors 276 may be connected to the Ethernet switch 206A through traces 280 on the switch board 272, and the connectors 278 may be connected to the storage device connectors 226 through traces 282 on the midplane 222E.

As with the other embodiments described above, the multi-channel cable assemblies may be connected at either end, with or without the use of connectors and/or conductive traces, using any suitable connection technique. However, because the cable assemblies may be connected between the switch boards and midplane without the high-speed connectors used in other embodiments, it may be beneficial to use a configuration that may be easily disconnected and reconnected at one or both ends. This may include, for example, the use of cable termination connectors that snap in to mating connectors and/or patterns of conductors on the circuit boards.

The embodiments disclosed above have been described in the context of various implementation details, but the principles of this disclosure are not limited these or any other specific details. For example, some functionality has been described as being implemented by certain components, but in other embodiments, the functionality may be distributed between different systems and components in different locations and having various user interfaces. Certain embodiments have been described as having specific processes, steps, etc., but these terms also encompass embodiments in which a specific process, step, etc. may be implemented with multiple processes, steps, etc., or in which multiple process, steps, etc. may be integrated into a single process, step, etc. A reference to a component or element may refer to only a portion of the component or element. For example, a reference to an integrated circuit may refer to all or only a portion of the integrated circuit, and a reference to a block may refer to the entire block or one or more subblocks. The use of terms such as “first” and “second” in this disclosure and the claims may only be for purposes of distinguishing the things they modify and may not to indicate any spatial or temporal order unless apparent otherwise from context. A reference to a first thing may not imply the existence of a second thing.

The various details and embodiments described above may be combined to produce additional embodiments according to the inventive principles of this patent disclosure. Since the inventive principles of this patent disclosure may be modified in arrangement and detail without departing from the inventive concepts, such changes and modifications are considered to fall within the scope of the following claims.

The invention claimed is:

1. An apparatus comprising:

a storage device comprising:

a first connector comprising a first multi-protocol storage device connector, wherein the first multi-protocol storage device connector comprises a first pin configured to transfer data using a first protocol and a second protocol; and

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a storage controller configured to control data transfers using the first multi-protocol storage device connector and a multi-level signaling protocol, wherein the multi-level signaling protocol uses multi-level pulse amplitude modulation; and

a circuit board comprising:

a second multi-protocol storage device connector, comprising a second pin configured to engage with the first pin and transfer data using the first protocol and the second protocol; and

a third connector having a configuration based on a performance characteristic comprising a linear transmission characteristic within a predetermined frequency range;

wherein the circuit board is configured to transfer data between the second multi-protocol storage device connector and the third connector.

2. The apparatus of claim 1, wherein the circuit board is configured to transfer data using the multi-level signaling protocol.

3. The apparatus of claim 1, wherein the circuit board comprises one or more traces configured to transfer data between the second multi-protocol storage device connector and the third connector.

4. The apparatus of claim 1, wherein the second multi-protocol storage device connector comprises a U.2 compatible connector.

5. The apparatus of claim 1, wherein the performance characteristic comprises a matched signal characteristic.

6. The apparatus of claim 1, wherein the multi-level signaling protocol uses a symbol to transfer a first bit of data and a second bit of data.

7. A system comprising:

a first circuit board configured to communicate using a multi-level signaling protocol; and

a second circuit board having a first multi-protocol storage device connector configured to couple the second circuit board to a storage device, wherein the first multi-protocol storage device connector comprises a first pin configured to transfer data using a first protocol and a second protocol;

wherein the multi-level signaling protocol uses multi-level pulse amplitude modulation;

wherein the second circuit board is configured to enable the storage device to communicate, using the first multi-protocol storage device connector, using the multi-level signaling protocol, with the first circuit board;

wherein the first multi-protocol storage device connector is configured to communicate with the first circuit board using a first cable and a second cable;

wherein the first cable has a first connector located on the second circuit board to communicate with the first multi-protocol storage device connector, and a second connector located on the first circuit board;

wherein the second cable has a third connector located on the second circuit board to communicate with a third multi-protocol storage device connector, and a fourth connector located on the first circuit board; and

wherein the storage device comprises:

a second multi-protocol storage device connector comprising a second pin configured to transfer data using the first protocol and the second protocol; and

a storage controller configured to control data transfers using the second multi-protocol storage device connector and the multi-level signaling protocol.

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8. The system of claim 7, wherein the second circuit board is coupled to the first circuit board using a circuit board connector.

9. The system of claim 7, wherein the second circuit board comprises one or more traces configured to connect the first connector of the first cable to the first multi-protocol storage device connector. 5

10. The system of claim 7, wherein the first connector of the first cable is physically coupled to the first multi-protocol storage device connector.

11. The system of claim 7, wherein the first multi-protocol storage device connector comprises a U.2 compatible connector. 10

12. A method comprising:

receiving, at a circuit board, using a multi-level signaling protocol, data; and 15

sending, from the circuit board, to a storage device, using the multi-level signaling protocol, the data;

wherein the multi-level signaling protocol uses multi-level pulse amplitude modulation;

wherein the receiving is performed using a first connector having a configuration based on a performance characteristic comprising a matched signal characteristic; 20

wherein the sending is performed using a second connector comprising a first multi-protocol storage device connector comprising a first pin configured to transfer data using a first protocol and a second protocol; and 25

wherein the storage device comprises:
a third connector comprising a second multi-protocol storage device connector comprising a second pin

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configured to transfer data using the first protocol and the second protocol; and

a storage controller configured to control data transfers using the second multi-protocol storage device connector and the multi-level signaling protocol.

13. The method of claim 12, wherein the multi-level signaling protocol uses pulse amplitude modulation.

14. The method of claim 13, wherein the pulse amplitude modulation comprises pulse amplitude modulation level-4 (PAM4) modulation. 10

15. The method of claim 12, wherein the multi-level signaling protocol comprises a network protocol.

16. The method of claim 12, wherein the receiving is performed using a non-volatile memory express over fabric (NVMe-oF) protocol. 15

17. The method of claim 12, wherein the first multi-protocol storage device connector comprises a U.2 compatible connector.

18. The method of claim 12, wherein the first multi-protocol storage device connector is compliant with a Small Form Factor 8639 (SFF-8639) specification.

19. The method of claim 12, wherein the performance characteristic comprises a linear transmission characteristic within a predetermined frequency range.

20. The method of claim 12, wherein the multi-level signaling protocol uses a symbol to transfer a first bit of data and a second bit of data.

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